COURIER MANAGEMENT SYSTEM

Salesforce-Based High-Level Project Report

Domain: Logistics, Courier & Delivery Operations

Platform: Salesforce Lightning / Salesforce Cloud

A comprehensive academic project document covering ideation, requirements, system architecture, shipment tracking, automation, security, analytics, testing and future scope.

PROJECT TEAM

Courier Management System – Salesforce-Based High-Level Project

The project team covers the major Salesforce implementation responsibilities across architecture, administration, business analysis, quality assurance, analytics and development.

Team Member	Role	Specialization
Sivasakthivel A	Salesforce Lead Developer & Architect	Custom Object Modeling, Salesforce Lightning Components, and Flow Automation
Saravanan M	Salesforce Admin & Security Specialist	Profiles, Permission Sets, Security Configurations, and Roll-up Summaries
Shyam R	Business Analyst & UI/UX Designer	Customer Journey Mapping, User Stories, and Lightning App Page Layouts
Sumith M	Quality Assurance & Analytics Engineer	Functional Testing, Reports, Dashboard Generation, and System Performance Validation
Vijayaprathap G	Salesforce Developer	Apex, SOQL, and Lightning Web Components

Team Responsibility Overview

The team structure distributes the project lifecycle across solution architecture and development, Salesforce administration and security, business analysis and user experience, quality assurance and analytics, and Apex/SOQL/Lightning Web Components development.

Project Platform: Salesforce Lightning / Salesforce Cloud

Project Domain: Logistics, Courier & Delivery Operations

CERTIFICATE

This is to certify that the project report entitled " **Courier Management System**" is a bonafide academic project prepared for the partial fulfilment of the requirements of the relevant programme. The project presents a Salesforce-based solution for managing courier bookings, customers, shipments, pickup and delivery operations, tracking, payments, notifications, reporting and role-based security.

Project Area	Coverage
Platform	Salesforce Lightning / Salesforce Cloud
Core Objects	Customer, Courier Shipment, Pickup, Hub, Delivery Agent, Tracking Event, Payment
Automation	Validation Rules, Record-Triggered Flows, Scheduled Flows, Notifications
Analytics	Reports, Dashboards, Operational KPIs
Security	Profiles, Permission Sets, Roles, Sharing Settings

DECLARATION

I declare that this project report is prepared for academic documentation and describes a proposed Salesforce-based Courier Management System. Institution-specific screenshots, actual deployment details, test execution results, student names and implementation evidence should be inserted before final submission.

ACKNOWLEDGEMENT

I express my sincere gratitude to the institution, project guide and faculty members for their guidance and support. I also acknowledge the supplied project guide, which provided the phase structure, requirements approach, architecture principles, implementation milestones, testing areas and future-scope framework used to organize this report.

PROJECT ABSTRACT

The Courier Management System (CMS) is a centralized cloud application designed to digitize and streamline courier booking and delivery operations. It maintains customer, shipment, pickup, hub, delivery-agent, tracking, payment and notification information in a unified Salesforce environment.

The system replaces fragmented registers, spreadsheets and manual status updates with structured records, validation, automated workflows, real-time tracking events, role-based access, reports and dashboards. A shipment can move through a controlled lifecycle from booking and pickup to hub processing, transit, out-for-delivery and final delivery.

The proposed solution improves operational visibility, data consistency, customer-service responsiveness and management reporting while providing a scalable foundation for integrations such as payment gateways, SMS/email providers, maps, barcode scanning and customer self-service portals.

KEYWORDS

Courier Management, Shipment Tracking, Logistics, Delivery Management, Salesforce Lightning, Custom Objects, Flow Automation, Validation Rules, Reports, Dashboards, Role-Based Security.

TABLE OF CONTENTS

Chapter	Topic
1	Introduction
2	Ideation Phase
3	Requirement Analysis Phase
4	Project Design Phase
5	Project Planning & Scheduling
6	Project Development / Salesforce Implementation
7	Functional & Performance Testing
8	Advantages, Limitations & Risks
9	Conclusion

10	Future Scope
11	Appendix – Configuration Reference

1. INTRODUCTION

1.1 Project Overview

The Courier Management System is a Salesforce application for managing end-to-end courier operations. It provides a single operational view of customers, shipments, addresses, pickup requests, hubs, delivery agents, tracking events, delivery outcomes and payments.

1.2 Purpose

- Digitize courier booking and shipment administration.
- Provide a reliable tracking history for every shipment.
- Automate repetitive operational tasks and notifications.
- Improve coordination between booking staff, hubs and delivery agents.
- Provide management with real-time reports and dashboards.
- Apply structured security and data-quality controls.

1.3 Objectives

- Create a single source of truth for courier records.
- Reduce duplicate and incomplete shipment information.
- Maintain chronological tracking events.
- Automate status transitions, alerts and follow-up activities.
- Enable role-based access to operational data.
- Support scalable cloud-based logistics operations.

1.4 Scope

In scope: customer registration, shipment booking, pickup scheduling, hub management, delivery-agent assignment, tracking events, delivery confirmation, payment/charge tracking, notifications, reports and dashboards.

Out of scope for the base academic version: live GPS navigation, external carrier marketplace integration, production payment settlement and large-scale route optimization; these are included as future extensions.

2. IDEATION PHASE

2.1 Problem Statement

User / Stakeholder	Current Problem	Operational Impact
Customer	Limited shipment visibility or delayed updates	Uncertainty and repeated enquiries

Booking Staff	Manual entry across registers/sheets	Duplicate or incomplete data
Hub Staff	Shipment movement tracked inconsistently	Sorting and dispatch delays
Delivery Agent	Assignments and delivery details scattered	Difficulty managing workload
Manager	Reports compiled manually	Delayed operational decisions
System Admin	Access managed inconsistently	Security and data-governance risk

2.2 Empathy Map

- Think & Feel: Customers want predictable delivery and clear tracking.
- See: Manual registers, spreadsheets, phone calls and fragmented status updates.
- Hear: Delivery enquiries, delay complaints and operational instructions.
- Say & Do: Book shipments, check tracking, call support and update delivery status.
- Pain: Delays, missing information, repeated data entry and unclear responsibility.
- Gain: Automated tracking, timely notifications and transparent shipment history.

2.3 Brainstorming & Selected Idea

Potential ideas included spreadsheet-based tracking, standalone web application, mobile-first courier application, barcode-driven tracking and a Salesforce-based logistics platform. The selected academic solution is a **Salesforce-based Courier Management System** because the platform supports structured data, relationships, security, automation, reports and dashboards in one environment.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey



Customer/Staff Login → Create Booking → Generate Tracking ID → Schedule Pickup → Hub Processing → Transit Updates → Assign Delivery Agent → Out for Delivery → Delivery Confirmation → Payment/Closure → Reporting.

3.2 Functional Requirements

ID	Requirement	Description
FR-01	User Authentication	Secure authenticated access for authorized users.
FR-02	Customer Management	Create, update and search customer profiles and addresses.
FR-03	Shipment Booking	Create shipment records with origin, destination, package and service details.
FR-04	Tracking	Maintain chronological shipment tracking events and current status.

FR-05	Pickup Management	Schedule and monitor pickup requests.
FR-06	Hub Operations	Record hub receipt, sorting, dispatch and transfer events.
FR-07	Delivery Assignment	Assign delivery agents and monitor delivery workload.
FR-08	Delivery Confirmation	Capture delivered/failed delivery outcome, date/time and proof reference.
FR-09	Payments	Record charges, payment status and transaction reference.
FR-10	Notifications	Trigger status-based customer/staff notifications.
FR-11	Reports	Generate filtered operational and management reports.
FR-12	Dashboards	Visualize shipments, delivery performance, exceptions and revenue.

3.3 Non-Functional Requirements

Attribute	Requirement
Usability	Simple Lightning navigation and role-specific pages.
Security	Profiles, permission sets, roles and sharing controls.
Reliability	Cloud-based centralized storage and controlled data entry.
Performance	Common record and report operations should respond promptly under expected academic workload.
Availability	Cloud access subject to Salesforce environment availability.
Scalability	Support additional customers, shipments, hubs, agents and regions.

4. PROJECT DESIGN PHASE

4.1 Problem- Solution Fit

The proposed system converts manual courier administration into a structured digital workflow. Centralized records improve visibility, validation reduces data-quality errors, automation reduces repetitive work, and dashboards provide a consolidated operational view.

4.2 Proposed Solution

Area	Design
Data	Salesforce custom objects for courier master and transaction records.
UI	Salesforce Lightning App with role-oriented navigation.
Quality	Required fields, picklists, validation rules and duplicate controls.
Relationships	Lookup and master-detail relationships where appropriate.
Automation	Record-triggered and scheduled flows; notifications and task creation.

Security	Profiles, permission sets, roles and sharing settings.
Analytics	Reports, dashboards and KPI monitoring.
Scalability	Cloud architecture with integration-ready design.

4.3 Solution Architecture



4.4 Data Flow

User → Lightning UI → Validation → Custom Objects → Automation / Tracking Events → Reports & Dashboards → Notifications / Operational Decisions.

4.5 Core Data Model

Object	Key Fields / Relationships
Customer	Customer Name, Phone, Email, Customer Type, Address; related to shipments.
Shipment	Tracking Number, Booking Date, Service Type, Origin, Destination, Weight, Charge, Current Status; related to Customer, Pickup, Tracking Events, Payment.

Object	Key Fields / Relationships
Pickup Request	Pickup Address, Requested Date/Time, Status, Assigned Staff/Agent; related to Shipment.
Hub	Hub Name, Code, Location, Contact; related to Tracking Events.
Delivery Agent	Agent Name, Phone, Employee ID, Zone, Availability; related to Delivery Assignments.
Tracking Event	Shipment, Event Type, Event Date/Time, Location/Hub, Remarks, Updated By.
Delivery Assignment	Shipment, Agent, Assigned Date, Delivery Status, Attempt Count.
Payment	Shipment, Amount, Payment Mode, Payment Status, Transaction Reference, Payment Date.

5. Project Planning & Scheduling

5.1 Development Plan

- Create and configure Salesforce development environment.
- Define objects, fields, relationships and record types.
- Build Lightning App pages and navigation.
- Configure profiles, permission sets, roles and sharing.
- Implement validation rules and duplicate controls.
- Develop Flow automations for booking, assignment, tracking and notifications.
- Create reports, dashboards and operational KPIs.
- Perform functional, integration-oriented and performance testing.
- Review documentation and prepare deployment/demo evidence.

5.2 Suggested 8-Week Schedule

Week	Activities	Deliverable
1	Ideation, stakeholder analysis, scope	Problem statement & requirements
2	Data model and UI design	ER/data model & page design
3	Objects, fields, relationships	Configured data layer
4	Lightning pages, forms, validation	Working user interface
5	Flows, tracking and notifications	Automated operational workflow
6	Security, reports, dashboards	Controlled analytics layer
7	Testing, defect fixing	Test evidence & stable build
8	Documentation, demo, final review	Final project package

5.3 Agile Work Items

Epic	Representative User Story
Booking	As a booking executive, I want to create a shipment so that a tracking number can be issued.
Tracking	As a customer-service user, I want to view shipment history so that I can answer status enquiries.
Delivery	As a manager, I want agents assigned to shipments so that deliveries can be monitored.
Notification	As a customer, I want status notifications so that I know when the shipment changes stage.
Analytics	As a manager, I want dashboards so that I can monitor delivery volume and exceptions.

6. Project Development / Salesforce Implementation

6.1 Salesforce Configuration

Component	Implementation
Custom Objects	Customer, Shipment, Pickup Request, Hub, Delivery Agent, Tracking Event, Delivery Assignment, Payment.
Fields	Text, Number, Currency, Date/Time, Picklist, Checkbox, Lookup/Master-Detail as required.
Record Types	Optional record types for Domestic, Express, Standard or Business shipment processes.
Lightning App	Navigation tabs for Customers, Shipments, Pickups, Hubs, Agents, Tracking, Payments and Reports.
Page Layouts	Role-specific layouts emphasizing required operational information.
Validation	Weight/charge/date/status consistency, mandatory destination and contact data.
Automation	Flows for tracking events, status updates, assignments, notifications and follow-up tasks.

6.2 Shipment Lifecycle Automation

Booking Created → Tracking Number Generated → Pickup Scheduled → Pickup Confirmed → Hub Received → In Transit → Destination Hub → Out for Delivery → Delivered / Delivery Exception.

Automation examples: when a shipment is created, validate mandatory data and create a tracking event; when pickup is confirmed, update shipment status; when an agent is assigned, create a delivery task; when delivery is confirmed, close the operational task and create a final tracking event; when an exception occurs, create an escalation task or notification.

6.3 Validation Rules

- Destination contact information must be present before booking confirmation.
- Package weight must be greater than zero.
- Expected delivery date cannot precede booking date.
- Delivered status requires delivery date/time and confirmation details.
- Payment amount cannot be negative.
- Tracking event time should not be earlier than the shipment booking time.

6.4 Notifications

Notifications may be generated for booking confirmation, pickup confirmation, in-transit milestones, out-for-delivery, successful delivery and delivery exceptions. Actual SMS/email integration can be connected through approved external services in a production implementation.

7. Functional & Performance Testing

7.1 Testing Strategy

Testing verifies that core business functions work as specified, invalid data is rejected, automation executes correctly, security boundaries are respected and reports reflect the underlying records. Production-like integrations should be tested separately before deployment.

TC ID	Test Scenario	Expected Result	Status*
TC-01	Valid customer creation	Customer record saved successfully	Pass / Pending
TC-02	Shipment without required destination	Validation prevents save/confirmation	Pass / Pending
TC-03	Create valid shipment	Tracking number/status workflow initiated	Pass / Pending
TC-04	Add tracking event	Event appears in shipment history	Pass / Pending
TC-05	Assign delivery agent	Assignment stored and follow-up created	Pass / Pending
TC-06	Mark shipment delivered without confirmation	System prevents invalid completion	Pass / Pending
TC-07	Record payment	Payment amount/status stored correctly	Pass / Pending
TC-08	Unauthorized data access	User cannot access restricted records	Pass / Pending
TC-09	Dashboard refresh	KPI reflects current test data	Pass / Pending
TC-10	Invalid weight / negative charge	Validation rejects invalid value	Pass / Pending

*Replace the status with actual execution evidence before submission.

7.2 Performance & Quality Areas

- Page and record navigation responsiveness.
- Report filtering and dashboard loading under expected academic data volume.
- Flow execution without recursive or duplicate updates.
- Data integrity across shipment and tracking relationships.
- Security validation using different user roles.
- Backup/export and recovery procedures appropriate to the project environment.

7.3 User Acceptance Criteria

The project is considered ready for academic demonstration when a user can create a customer and shipment, move the shipment through representative lifecycle stages, view tracking history, assign a delivery agent, record delivery/payment information, and generate management reports without data-quality or authorization errors.

8. Advantages, Limitations & Risks

8.1 Advantages

- Centralized courier and customer information.
- End-to-end shipment visibility through tracking events.
- Reduced manual data entry through automation.
- Consistent validation and controlled status progression.
- Role-based access improves operational data governance.
- Reports and dashboards support faster monitoring.
- Cloud platform provides an extensible foundation for integrations.

8.2 Limitations

- Live GPS tracking is not included in the base academic build.
- External SMS/email/payment services require separate configuration.
- Internet/cloud availability affects access.
- Advanced route optimization requires additional algorithms or services.
- Production-scale compliance, audit and data-retention policies need organization-specific design.

8.3 Risks & Mitigations

Risk	Impact	Mitigation
Incorrect shipment data	Tracking/fulfilment errors	Required fields, validation and review.
Unauthorized access	Privacy/security exposure	Profiles, permission sets, roles and sharing.
Automation failure	Delayed updates	Flow testing, fault handling and monitoring.
Duplicate records	Confused customer/shipment history	Duplicate rules and unique tracking number.
Integration outage	Notifications/payments delayed	Queue/retry strategy and exception logging.

8.4 Business Impact

The proposed solution is intended to improve operational transparency, reduce repetitive administrative effort, support faster customer-service responses and provide management with consistent shipment and delivery metrics.

9. Conclusion

The Courier Management System provides a structured digital approach to managing courier operations from booking through final delivery. By combining Salesforce Lightning interfaces, custom objects, validation rules, Flow automation, role-based security, tracking events and analytics, the solution addresses common problems caused by fragmented manual processes.

The architecture is intentionally modular: customer and shipment data form the operational core; pickup, hub, delivery-agent and tracking objects support execution; payment and notification components extend the lifecycle; reports and dashboards provide management visibility.

For an academic implementation, the project demonstrates practical application of CRM/cloud concepts, data modeling, automation, security, user-interface design, testing and analytics in a logistics domain. Before production use, organization-specific security, integration, compliance, service-level and operational requirements should be validated.

Key Project Outcomes

Outcome Area	Expected Result
Data Management	Centralized and structured courier records
Tracking	Chronological shipment status visibility
Automation	Reduced repetitive operational tasks
Security	Controlled access based on user responsibilities
Analytics	Operational KPIs through reports and dashboards
Scalability	Foundation for mobile, integration and advanced analytics extensions

10. Future Scope

The base project can be expanded into a more advanced logistics platform through the following extensions:

Future Feature	Description
Customer Self-Service Portal	Customers can book shipments, view tracking and download receipts.
Mobile Application	Dedicated mobile workflow for delivery agents and field operations.
GPS / Map Integration	Live location, route visualization and delivery-zone mapping.
Barcode / QR Scanning	Fast shipment identification at pickup, hub and delivery points.
Payment Gateway	Online payment collection, reconciliation and digital receipts.
SMS / WhatsApp / Email Integration	Automated milestone notifications and customer communication.
AI / Predictive Analytics	Delivery-delay risk, volume forecasting and exception prioritization.
Route Optimization	Suggest delivery sequences based on location, time and workload.
Partner / Carrier Integration	Exchange shipment events with external logistics providers through APIs.

Advanced SLA Monitoring	Measure promised-vs-actual delivery time and escalation thresholds.

10.1 Long-Term Vision

The long-term vision is a connected logistics ecosystem where shipment information flows automatically across customers, booking teams, hubs, delivery agents, payment services and external carrier systems while management receives near-real-time operational intelligence.

11. APPENDIX – CONFIGURATION REFERENCE

11.1 Suggested Object Inventory

Object	Type / Purpose	Important Fields
Customer	Master data	Name, Phone, Email, Type, Billing/Delivery Address
Shipment	Core transaction	Tracking No., Customer, Service, Origin, Destination, Weight, Charge, Status
Pickup Request	Operational transaction	Shipment, Address, Date/Time, Status, Assigned Agent
Hub	Master data	Name, Code, Location, Contact
Delivery Agent	Master data	Name, Employee ID, Phone, Zone, Availability
Tracking Event	Transaction history	Shipment, Event Type, Date/Time, Location, Remarks
Delivery Assignment	Operational transaction	Shipment, Agent, Assigned Date, Status, Attempts
Payment	Financial record	Shipment, Amount, Mode, Status, Transaction Ref, Date

11.2 Suggested Roles

Role	Primary Responsibilities
System Administrator	Configuration, users, security, automation and governance.
Operations Manager	Monitor shipments, hubs, agents, exceptions and KPIs.
Booking Executive	Create customers, bookings, pickup requests and basic updates.
Hub Operator	Receive, sort, transfer and update hub-related tracking events.
Delivery Agent	View assigned deliveries and update delivery outcomes.
Customer Service	Search customers/shipments and respond to tracking enquiries.
Finance User	Manage payment/charge records and financial reports.

11.3 Suggested Dashboard KPIs

- Total shipments booked today / this week / this month.
- Shipments by current status.
- Delivered versus pending shipments.
- Delivery exception count.
- Average delivery duration.

Report	Purpose
Shipment Status Summary	Monitor shipment volume by status.
Daily Booking Report	Review daily booking activity.

- Agent workload by zone.
- Hub-wise shipment volume.
- Pending payment amount and payment status.
- Service-type distribution (Standard / Express etc.).

11.4 Sample Report Catalogue

Report	Purpose
Pending Delivery Report	Identify shipments requiring delivery action.
Exception Report	Track failed/held/delayed shipments.
Agent Performance Report	Compare assigned and completed delivery workload.
Hub Volume Report	Monitor inbound/outbound volume by hub.
Payment Summary	Review charges and payment status.

11.5 Academic Submission Checklist

- Insert institution logo and approved certificate format.
- Add student names, register numbers, department and guide details.
- Insert screenshots of Salesforce objects, Lightning pages, flows, reports and dashboards.
- Replace sample test statuses with actual executed test results.
- Attach ER/data model, architecture and workflow diagrams if required.
- Verify formatting and page numbering against department guidelines.
- Include actual deployment/demo evidence where applicable.

PROJECT SUMMARY

Project Title: Courier Management System

Platform: Salesforce Lightning / Salesforce Cloud

Domain: Courier, Logistics & Delivery Management

Core Capability: Customer + Shipment + Pickup + Hub + Agent + Tracking + Delivery + Payment + Analytics **Primary**

Outcome: A centralized, automated and secure courier operations platform.

END OF PROJECT REPORT

Prepared as a high-level academic project document based on the supplied guide structure.